

EDUCATION

Tulane University, School of Science and Engineering
B.S. in Mathematics, GPA: 3.97 / 4.00

New Orleans, LA
August 2024 – May 2027, Expected

RESEARCH INTERESTS

Post-quantum and code-based cryptography for resource-constrained devices; algebraic coding theory; hardware-assisted key establishment; quantum information theory.

PUBLICATIONS & PREPRINTS

- [1] M. B. Can and E. C. Naig, “A Theory of Leakage in Secret Sharing and Evaluation Codes”, Submitted, under review, 2026.
- [2] M. B. Can and E. C. Naig, *Applications of Bruhat-Chevalley-Renner Decomposition to Metric-Aware Code-Based Cryptography*, Cryptology ePrint Archive, Paper 2026/665, Preprint, 2026. [Online]. Available: <https://eprint.iacr.org/2026/665>
- [3] M. B. Can and E. C. Naig, “Module-HQC”, Submitted to IEEE Transactions on Information Theory, 2026.
- [4] E. C. Naig and S. A. Fenner, *Parallel Repetition in the Two-Player Quantum Cloning Game*, arXiv:2608.00930 [quant-ph], Preprint, 2026. [Online]. Available: <https://arxiv.org/abs/2608.00930>

TALKS & PRESENTATIONS

- *Parallel Repetition in the Two-Player Quantum Cloning Game*. Talk presented by co-author S. A. Fenner, Ulm University, Ulm, Germany ([event page](#)). September 2026
- *Quantum Position Verification and the Quantum Cloning Game*. Poster, USC Summer Research Symposium, University of South Carolina. July 2026
- *Quantum Position Verification: Cloning Games and an Open Repetition Problem*. Talk to faculty and students of the NSF REU in Quantum Information Science, University of South Carolina. July 2026

PATENTS

- **Tunable Module Hamming Quasi-Cyclic Code-Based Cryptography**. U.S. provisional patent application, with M. B. Can. Filed August 2026
- **Systems and Methods for Secure and Reliable Data Storage and Transmission**. PCT patent application, with M. B. Can and D. Montero. Filed April 2026
- **Metric-Aware Code-Based Cryptography**. U.S. provisional patent application, with M. B. Can. Filed March 2026

RESEARCH EXPERIENCE

Undergraduate Researcher

Advisor: Mahir Bilen Can, Tulane University

New Orleans, LA

August 2025 – Present

NSF REU in Quantum Information Science

Advisor: Stephen Fenner, University of South Carolina

Columbia, SC

May 2026 – August 2026

RTG Summer Math Program

Louisiana State University

Baton Rouge, LA

May 2025 – June 2025

TECHNICAL & ENTREPRENEURIAL EXPERIENCE

QSPARC Labs

Co-Founder, CEO, and Lead Developer

New Orleans, LA

August 2025 – Present

TEACHING & MENTORING

- Volunteer Tutor, Excite All Stars Fall 2026 – Present
- Volunteer, College Beyond Fall 2026 – Present
- Volunteer Tutor, Tulane University Fall 2025 – Present

AWARDS & GRANTS

- Tulane Honors Scholar 2024–Present
- Dean's List, Tulane University Fall 2024; Spring 2025; Fall 2025; Spring 2026
- CELT Faculty-Mentored Undergraduate Research Grant, Tulane University; mentor: Mahir Bilen Can. 2026
- Chosen for a Tulane Innovation Institute program on research translation and proposal development. 2025
- Ellen Applebaum Temkin Scholarship, Mathematics Department, West High School 2024

LEADERSHIP & SERVICE

- President, Tulane Math Club Spring 2026–Present
- Co-Founder and Vice President, Tulane Science Olympiad 2024–Present
- Trip Leader, Tulane Outdoor Adventures 2024–Present